A Robust Adversarial Defense Algorithm for Enhancing Salient Object Detection in Remote Sensing Image

Xueli Shi, Zhi Li, Yi Wang, Yu Lu, and Li Zhang

***Abstract*— Deep neural networks (DNNs) have been widely applied to salient object detection in optical remote sensing images (ORSIs-SOD) with significant progress. However, the introduction of adversarial perturbations can severely degrade their detection performance. Moreover, the relatively limited labeled samples in ORSI-SOD make the model more susceptible for overfitting, increasing sensitivity to adversarial perturbations. Therefore, researching defense algorithms against adversarial samples to enhance the robustness of salient object detection (SOD) models in the field of ORSIs is of great importance. However, existing defense studies have seldom focused on the ORSI-SOD task and have not considered the unique challenges it faces to propose effective defense solutions. Thus, this article proposes a robust adversarial defense algorithm specifically for ORSI-SOD for the first time. This algorithm first proposes a fixed luminance channel multimodal data random fusion adversarial defense strategy to enhance the model's robustness against the color and contour information of targets. Then, the algorithm proposes a sinusoidal noise-driven adversarial feature adaptive masking module, which adaptively reduces the impact of adversarial perturbations and enhances the model's generalization ability. In addition, the algorithm proposes a global context-aware denoising perception module (GCA-DPM) into the network to strengthen the model's understanding of the global content of images, further suppressing the impact of adversarial noise and enhancing the detection accuracy. Finally, experimental results demonstrate that the proposed defense algorithm effectively enhances the adversarial robustness of various SOD models. Against three different attack strategies, the algorithm increases the average F_β of four SOD models by 23.94% and 21.98% on the optical remote sensing saliency detection (ORSSD) and extended ORSSD (EORSSD) datasets, respectively.**

***Index Terms*— Adversarial defense, remote sensing images, salient object detection (SOD).**

I. Introduction

SALIENT object detection (SOD) [1] is a method that simulates human visual attention mechanisms. It assists machines to automatically identify the most attractive regions or objects within an image. As a crucial preprocessing step, SOD has driven advancements in fields, such as object segmentation [2], image quality assessment [3], and object tracking [4]. In recent years, convolutional neural networks (CNNs), as a widely used deep learning architecture in the field of computer vision, have provided strong support for the further development of SOD tasks.

In natural scene images (NSIs), CNN-based SOD tasks have demonstrated excellent detection performance and achieved many significant research results [5], [6]. SOD also holds significant importance and practical value in the field of ORSIs [7]. However, detection in ORSIs faces more challenges compared with NSIs. First, NSIs have relatively simple backgrounds with high contrast between objects and backgrounds, making target differentiation easier. In contrast, ORSIs typically contain complex and diverse geographical features, increasing the difficulty of target separation. Second, NSIs are usually captured at closer distances with manually adjustable equipment, which aids in obtaining images with clear edge features, and the variation in target sizes is relatively limited. However, ORSIs are often captured from high altitudes or space, where targets in the images may have blurred edge features due to the atmospheric scattering and cloud cover. In addition, varying shooting angles result in significant differences in target sizes, especially for small targets, making it more difficult to extract effective features and thus increasing the complexity of detection. Finally, due to the high cost and complexity of acquiring and annotating remote sensing images, ORSIs face significant limitations in terms of data volume and diversity compared with NSIs [8]. This may result in insufficient generalization ability of the model and a higher risk of overfitting when dealing with ORSIs [9]. To specifically enhance the performance of SOD in optical remote sensing image (ORSI-SOD) tasks, numerous CNN-based solutions [10], [11] have been proposed. These methods, by deeply exploring the potential of CNNs, have achieved more efficient and accurate detection of objects in the complex scenes of ORSIs.

However, studies [12], [13], [14] have already demonstrated the vulnerability of deep learning-based SOD networks when confronted with adversarial attacks. Even the slightest perturbations can lead to a significant decrease in the network's detection performance. Consequently, the development of attack and defense algorithms based on adversarial samples

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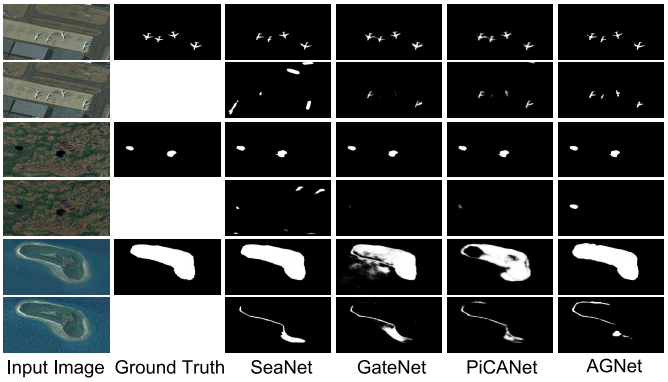


Fig. 1. Model sensitivity to adversarial attacks. The leftmost column displays the input images, with natural samples shown above and their corresponding PGD adversarial samples shown below. The second column presents the ground-truth saliency maps. Since natural images and their corresponding adversarial samples share the same ground truth, the lower areas of this column are left blank. As depicted, when the input images are subjected to adversarial perturbations, various detection models exhibit varying degrees of decreased saliency map quality.

has become a crucial research topic in deep learning [15], [16]. In the field of classification of remote sensing images, the issue of adversarial samples has been studied in depth [17], [18]. For example, Lu et al. [19], [20] combined the proactive and reactive defense strategies to enhance the robustness of deep ensemble models, addressing the adversarial challenges faced by unmanned aerial vehicles in remote sensing image classification recognition. Xu et al. [21], [22] improved the adversarial robustness of remote sensing image classification models by combining adversarial training and self-supervised learning methods without relying on large amounts of labeled data. Su et al. [23] proposed an adversarial training framework that combines reconstruction assistance and distance optimization to enhance the defense capability of deep neural networks (DNNs) against adversarial samples, addressing the issues of interclass similarity and scale variation in remote sensing image classification. In the field of object detection in remote sensing images [24], research on adversarial samples is also gradually gaining attention. For example, recent works [25], [26] effectively disrupted detection systems for remote sensing images by altering image background information in the physical world. However, research on adversarial samples in the context of saliency detection in remote sensing images remains relatively scarce, particularly with no studies having proposed effective defense solutions for the challenges faced by this task. Saliency detection plays a pivotal role in processing remote sensing images, aiming to extract important objects, such as buildings, roads, and vehicles from these images. In adversarial attacks, adding a small proportion of noise to natural samples can cause large differences in saliency predictions between adversarial and natural samples. As illustrated in Fig. 1, even though adversarial and natural samples have minimal visual differences, advanced SOD networks exhibit noticeable deviations between the detection results and the ground truth when applied to adversarial samples, demonstrating severe security risks. Therefore, conducting research on defense mechanisms against adversarial samples in the context of saliency detection tasks in optical remote sensing holds significant theoretical importance and practical value.

Color features in remote sensing images can be used to differentiate various land features, objects, and scenes in SOD tasks. However, due to factors, such as varying lighting conditions, the diversity of targets, differences in color distributions, and limitations in the quality of remote sensing data, color features are susceptible to interference in prediction tasks [27], [28], [29]. Moreover, if a model relies too heavily on color features for detection, it may fail to effectively distinguish between objects with similar color and the background, reducing the detection accuracy for specific targets [30]. Recent research [31] indicates that in the YCbCr color space, adversarial noise in the luminance channel is particularly disruptive to models. Furthermore, under fixed perturbation conditions, the addition of perturbations to the luminance channel results in a significant decrease in the model's classification success rate. In addition, in ORSI-SOD, the relatively limited labeled samples increase the risk of model overfitting, leading to heightened sensitivity to noise and making the model more susceptible to adversarial perturbations [32]. Therefore, it is necessary to enhance the model's generalization ability to reduce its vulnerability to adversarial perturbations [33]. Inspired by above-mentioned studies, this article proposes a robust adversarial defense algorithm, for SOD tasks in optical remote sensing imagery. First, we convert input images from RGB to YCbCr color space. Considering the crucial role of shape and contour features in enhancing the robustness of defense mechanisms [34], sketch images are introduced during model training to strengthen the representation of morphological information. In this process, we propose an adversarial defense strategy, which fixes the luminance channel, randomly discards color channels, and then fuses the corresponding sketch image with the remote sensing image to generate a new trichannel input. This adversarial defense strategy enhances the model's robustness to color variations and contour information of targets, thereby reducing interference from complex backgrounds. Second, considering that adding robust noise to the input or internal layers of the model can enhance its generalization ability and reduce the effectiveness of adversarial attacks [13], [14], we propose a sinusoidal noise-driven adversarial feature adaptive masking module. This module not only adaptively hides adversarial sample features by dynamically adjusting noise parameters in the sinusoidal space but also uses the randomness of this noise to expose the model to a broader sample distribution, thereby reducing the model's sensitivity to adversarial perturbations. Finally, considering the small size of targets and the blurriness of edge features in ORSIs, which increase the difficulty of target recognition for the model [35], we further propose a global context-aware denoising perception module. It is embedded after each main convolutional block in the encoder of the SOD network. This module uses a feature fusion strategy based on the context-aware mechanisms and channel attention to enhance the saliency of target features and strengthen the model's understanding of the global content of the image. This allows the model to improve its robustness against adversarial noise and improve the accuracy of SOD even in the presence of noise. The proposed defense algorithm can be integrated as a plugin into different SOD networks without requiring

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TABLE I
TRAINING PARAMETERS OF EACH SOD MODEL

Hyper-parameters	SeaNet	GateNet	PiCANet	AGNet
Epochs	50	40	80	60
Batch Size	8	4	10	8
Optimizer	Adam	SGD	SGD	Adam
Learning Rate	1e-4	1e-3	1e-3	1e-4
Momentum	-	0.9	0.9	-
Betas	(0.9,0.999)	-	-	(0.9,0.999)
Weight Decay	0	5e-4	5e-4	5e-4
Image Size	288×288	384×384	224×224	224×224

adjustments to the already deployed model structures. The main contributions of this article can be summarized as follows.

1) We propose a robust adversarial defense algorithm designed for the ORSI-SOD task for the first time. This algorithm enhances the robustness of SOD networks against adversarial samples and can be easily integrated as a plugin into various SOD networks without requiring modifications to the original model structure.
2) We propose a fixed luminance channel multimodal data random fusion adversarial defense strategy. This strategy effectively enhances the model's ability to recognize color variations and contour features of salient objects in remote sensing images in the interference of complex backgrounds.
3) We propose a sinusoidal noise-driven adversarial feature adaptive masking module and a global context-aware denoising perception module. The former is designed to adaptively reduce the interference of adversarial samples and improve the model's generalization ability. The latter aims to strengthen the model's understanding of the global content in images to further mitigate the negative impact of adversarial noise and enhance the accuracy of SOD.
4) Extensive experiments demonstrate that the proposed defense algorithm effectively enhances the adversarial robustness of various SOD models. Against three different attack strategies, the algorithm increases the average F_β of four SOD models by 23.94% and 21.98% on the optical remote sensing saliency detection (ORSSD) and extended ORSSD (EORSSD) datasets, respectively.

II. RELATED WORKS

A. Salient Object Detection for Optical Remote Sensing

In the field of optical remote sensing, the SOD technology plays a vital role in analyzing and understanding physical scenes within remote sensing images. With the construction of related datasets [8], [36], CNN-based methods for ORSI-SOD have rapidly advanced [10], [11]. These methods have been extensively applied in various domains, such as agriculture, forestry, environmental science, and security monitoring, including practical applications such as building recognition [37], airport monitoring [38], oil tank detection [39], and ship detection [40]. However, despite the significant achievements of these deep learning-based methods in improving the performance of ORSI-SOD, they still exhibit vulnerability when faced with adversarial samples. Adversarial samples are those that introduce subtle, nearly imperceptible perturbations into natural samples, resulting in a decrease in model performance. Even though these samples may have minimal visual differences from natural samples, they can significantly impact the neural network's detection performance, leading to severe misjudgments. Thus, unlike previous research focused on enhancing the performance of ORSI-SOD, this article places emphasis on defense strategies for deep learning-based SOD models in the optical remote sensing domain when dealing with adversarial samples. Our aim is to enhance the robustness of SOD models against adversarial examples, thereby increasing the stability and reliability of deep learning-based ORSI-SOD methods in practical applications.

B. Adversarial Attack

Adversarial attack strategies can be categorized into gradient-based and gradient-free attacks. Gradient-based attacks are more powerful and typically computationally less expensive, making them more meaningful to defend against. This article utilizes the following three typical gradient-based attack methods to evaluate the effectiveness of the proposed defense algorithm.

1) Fast Gradient Sign Method [41]: It is an adversarial attack method based on a single-step optimization in the direction of the gradient. For a given natural sample x and its corresponding target label y, fast gradient sign method (FGSM) modifies each element along the gradient of the loss function $L(x, y, \theta)$ with respect to the input x, where θ denotes the network parameters, and the hyperparameter ϵ controls the maximum perturbation range between the generated adversarial sample x' and the natural sample x. The process of generating the FGSM adversarial sample x' can be expressed as follows:

$$x' = x + \epsilon \mathrm{sgn}(\nabla_x L(x, y, \theta)). \tag{1}$$

2) Projected Gradient Descent (PGD) Method [42]: It is an adversarial attack method based on multiple iterative optimizations using FGSM as a basis. It is considered as one of the most powerful adversarial attacks and is commonly used to assess the robustness of adversarial defense algorithms. In generating adversarial samples, the original input x is used as the initial value, and adversarial sample x' is generated through multiple steps of gradient iterations. In the tth step of the iteration, the process of generating the adversarial sample x' can be expressed as follows:

$$x'_t = \Pi_{P_\epsilon(x)}\left(x'_{t-1} + a \cdot \mathrm{sgn}\left(\nabla_x L\left(g\left(x'_{t-1}; \theta\right), y\right)\right)\right) \tag{2}$$

where Π represents the projection operation, $P_\epsilon(x)$ is the projection space under the constraint of $x \pm \epsilon$, and a is the step size.

3) Automatic Projected Gradient Descent (APGD) [43]: This method improves upon the PGD approach. It optimizes the generation process of adversarial samples by automatically adjusting the attack step size and incorporating a momentum term during the iteration process. Let x be the initial input, given an initial step size a and a number of iterations N.

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TABLE II
PERFORMANCE OF SOD MODELS UNDER ADVERSARIAL ATTACKS

Dataset	Attack	SeaNet			GateNet			PiCANet			AGNet		
		$F_\beta \uparrow$	MAE↓	$S_m \uparrow$	$F_\beta \uparrow$	MAE↓	$S_m \uparrow$	$F_\beta \uparrow$	MAE↓	$S_m \uparrow$	$F_\beta \uparrow$	MAE↓	$S_m \uparrow$
ORSSD	Clean Image	0.8942	0.0105	0.9260	0.8899	0.0118	0.9211	0.8732	0.0199	0.8891	0.9123	0.0091	0.9395
	FGSM	0.6548	0.0490	0.7215	0.7536	0.0439	0.7594	0.6932	0.0819	0.7287	0.7512	0.0516	0.8099
	PGD	0.2051	0.1244	0.4705	0.6521	0.0912	0.7101	0.6199	0.0793	0.6566	0.6799	0.0599	0.7833
	APGD	0.1899	0.1436	0.4656	0.7178	0.0629	0.7366	0.6810	0.0832	0.7125	0.7247	0.0583	0.7856
EORSSD	Clean Image	0.8649	0.0073	0.9208	0.8698	0.0086	0.8737	0.8543	0.0130	0.8691	0.8891	0.0072	0.9285
	FGSM	0.6296	0.0472	0.7137	0.7421	0.0321	0.7502	0.6821	0.0421	0.7368	0.7399	0.0283	0.7952
	PGD	0.1496	0.0916	0.4908	0.6321	0.0895	0.6920	0.5995	0.0641	0.6211	0.6694	0.0324	0.7472
	APGD	0.1302	0.1376	0.4616	0.6944	0.0599	0.7279	0.6560	0.0558	0.6947	0.7086	0.0304	0.7583

TABLE III
PERFORMANCE COMPARISON OF DEFENSE ALGORITHMS ON FGSM, PGD, AND APGD ADVERSARIAL SAMPLES

Attack	Dataset	Method	SeaNet			GateNet			PiCANet			AGNet		
			$F_\beta \uparrow$	MAE↓	$S_m \uparrow$	$F_\beta \uparrow$	MAE↓	$S_m \uparrow$	$F_\beta \uparrow$	MAE↓	$S_m \uparrow$	$F_\beta \uparrow$	MAE↓	$S_m \uparrow$
FGSM	ORSSD	No Defense	0.6548	0.0402	0.7215	0.7536	0.0439	0.7594	0.6932	0.0819	0.7287	0.7512	0.0516	0.8099
		Smooth	0.7426	0.0439	0.7699	0.8133	0.0314	0.8201	0.7771	0.0430	0.7950	0.8593	0.0170	0.8870
		JPEG	0.7815	0.0330	0.8156	0.8259	0.0267	0.8411	0.8220	0.0286	0.8339	**0.8895**	**0.0140**	0.9063
		NRP	0.7805	0.0339	0.8086	0.8096	0.0306	0.8369	0.8110	0.0315	0.8358	0.8677	0.0170	0.8953
		RAN	0.7662	0.0351	0.7915	0.8023	0.0317	0.8295	0.7944	0.0333	0.8275	0.8598	0.0182	0.8898
		LeNo	0.7789	0.0317	0.8104	0.8225	0.0261	0.8498	0.8197	0.0272	0.8371	0.8697	0.0161	0.9009
		Ours	**0.8223**	**0.0264**	**0.8460**	**0.8526**	**0.0228**	**0.8678**	**0.8309**	**0.0244**	**0.8435**	0.8851	0.0141	**0.9075**
	EORSSD	No Defense	0.6296	0.0473	0.7137	0.7421	0.0321	0.7502	0.6821	0.0421	0.7368	0.7399	0.0283	0.7952
		Smooth	0.6887	0.0273	0.7461	0.7774	0.0178	0.7844	0.7380	0.0258	0.7666	0.7948	0.0113	0.8515
		JPEG	0.7364	0.0208	0.7911	0.8014	0.0136	0.8225	**0.7934**	0.0163	0.8131	0.8295	0.0092	0.8612
		NRP	0.7216	0.0207	0.7805	0.7822	0.0149	0.8123	0.7726	0.0184	0.8042	0.8217	0.0100	0.8638
		RAN	0.7149	0.0223	0.7709	0.7780	0.0159	0.8003	0.7685	0.0198	0.7939	0.8107	0.0111	0.8558
		LeNo	0.7266	0.0203	0.7845	0.7936	0.0132	0.8233	0.7878	0.0160	0.8091	0.8244	0.0094	0.8672
		Ours	**0.7738**	**0.0171**	**0.8128**	**0.8115**	**0.0129**	**0.8265**	0.7917	**0.0156**	**0.8158**	**0.8371**	**0.0090**	**0.8705**
PGD	ORSSD	No Defense	0.2051	0.1244	0.4705	0.6521	0.0633	0.7101	0.6199	0.0793	0.6566	0.6799	0.0599	0.7833
		Smooth	0.6778	0.0493	0.7224	0.7747	0.0360	0.7815	0.7702	0.0599	0.7793	0.8010	0.0382	0.8365
		JPEG	0.7732	0.0335	0.8070	0.8231	0.0280	0.8390	0.8016	0.0355	0.8265	0.8778	**0.0150**	**0.9041**
		NRP	0.7705	0.0342	0.8016	0.8262	0.0292	0.8367	0.7955	0.0317	0.8264	0.8641	0.0164	0.8672
		RAN	0.7454	0.0366	0.7808	0.8159	0.0303	0.8277	0.7816	0.0325	0.8005	0.8577	0.0173	0.8605
		LeNo	0.7687	0.0327	0.8005	0.8308	0.0272	0.8550	0.8011	0.0316	0.8268	0.8686	0.0161	0.8744
		Ours	**0.8320**	**0.0252**	**0.8546**	**0.8647**	**0.0206**	**0.8791**	**0.8288**	**0.0272**	**0.8479**	**0.8815**	0.0160	0.8970
	EORSSD	No Defense	0.1496	0.0916	0.4908	0.6321	0.0895	0.6920	0.5995	0.0641	0.6211	0.6694	0.0324	0.7472
		Smooth	0.6349	0.0296	0.7088	0.7455	0.0227	0.7601	0.7195	0.0438	0.7299	0.7838	0.0193	0.8312
		JPEG	0.7402	0.0205	0.7893	0.7970	0.0143	0.8115	0.7752	0.0206	0.7998	0.8255	0.0099	**0.8670**
		NRP	0.7134	0.0208	0.7729	0.7865	0.0148	0.8036	0.7628	0.0182	0.8014	0.8110	0.0106	0.8570
		RAN	0.6925	0.0233	0.7587	0.7792	0.0158	0.7979	0.7575	0.0201	0.7868	0.8092	0.0112	0.8408
		LeNo	0.7371	0.0194	0.7812	0.7912	0.0133	0.8214	0.7729	0.0173	0.8002	0.8185	0.0102	0.8621
		Ours	**0.7726**	**0.0157**	**0.8153**	**0.8173**	**0.0116**	**0.8346**	**0.7819**	**0.0167**	**0.8131**	**0.8317**	**0.0098**	0.8663
APGD	ORSSD	No Defense	0.1899	0.1436	0.4656	0.7178	0.0629	0.7366	0.6810	0.0832	0.7125	0.7247	0.0583	0.7856
		Smooth	0.6968	0.0478	0.7430	0.7914	0.0347	0.8101	0.7619	0.0459	0.7892	0.8184	0.0371	0.8415
		JPEG	0.7947	0.0323	0.8284	0.8514	0.0241	0.8751	0.8064	0.0347	0.8275	0.8882	0.0137	0.9097
		NRP	0.7784	0.0340	0.8036	0.8149	0.0304	0.8458	0.7991	0.0352	0.8190	0.8559	0.0179	0.8727
		RAN	0.7667	0.0350	0.7990	0.8122	0.0307	0.8406	0.7863	0.0337	0.8086	0.8491	0.0188	0.8655
		LeNo	0.7906	0.0306	0.8231	0.8482	0.0239	0.8669	0.8051	0.0314	0.8294	0.8824	0.0138	0.9057
		Ours	**0.8221**	**0.0293**	**0.8331**	**0.8758**	**0.0173**	**0.8872**	**0.8111**	**0.0293**	**0.8355**	**0.8895**	**0.0135**	**0.9104**
	EORSSD	No Defense	0.1302	0.1376	0.4616	0.6944	0.0599	0.7279	0.6560	0.0558	0.6947	0.7086	0.0304	0.7583
		Smooth	0.6599	0.0282	0.7217	0.7515	0.0212	0.7766	0.7466	0.0248	0.7637	0.7909	0.0115	0.8443
		JPEG	0.7462	0.0203	0.7905	0.8084	0.0134	0.8279	**0.7938**	**0.0162**	0.8191	0.8362	0.0091	0.8696
		NRP	0.7312	0.0201	0.7870	0.7722	0.0146	0.8161	0.7755	0.0181	0.7996	0.8278	0.0097	0.8689
		RAN	0.7301	0.0214	0.7741	0.7688	0.0166	0.8079	0.7659	0.0203	0.7784	0.8181	0.0109	0.8587
		LeNo	0.7587	0.0185	0.8039	0.8174	0.0125	0.8307	0.7785	0.0171	0.8058	0.8316	0.0093	0.8665
		Ours	**0.7812**	**0.0170**	**0.8270**	**0.8359**	**0.0103**	**0.8483**	0.7911	0.0164	**0.8262**	**0.8452**	**0.0088**	**0.8773**

The process of generating the adversarial sample x' can be formulated as follows:

$$\begin{aligned} z_t &= \Pi_{P_\epsilon}(x)\left(x'_{t-1} + a_{t-1} \cdot \nabla_x L\left(g\left(x'_{t-1};\theta\right), y\right)\right) \\ x'_t &= \Pi_{P_\epsilon}(x)\left(x'_{t-1} + \eta\left(z_t - x'_{t-1}\right) + (1-\eta)\left(x'_{t-1} - x'_{t-2}\right)\right) \end{aligned} \tag{3}$$

where the intermediate variable z_t is used to update the adversarial sample at the tth iteration, a_{t-1} represents the step size from the previous iteration, η is the weight of the momentum term, typically set to 0.75. The algorithm stops running after performing N iterations.

C. Adversarial Defense

Adversarial defense strategies are mainly divided into three categories: adversarial training, improving model structure, and input transformation-based defense strategies. Adversarial

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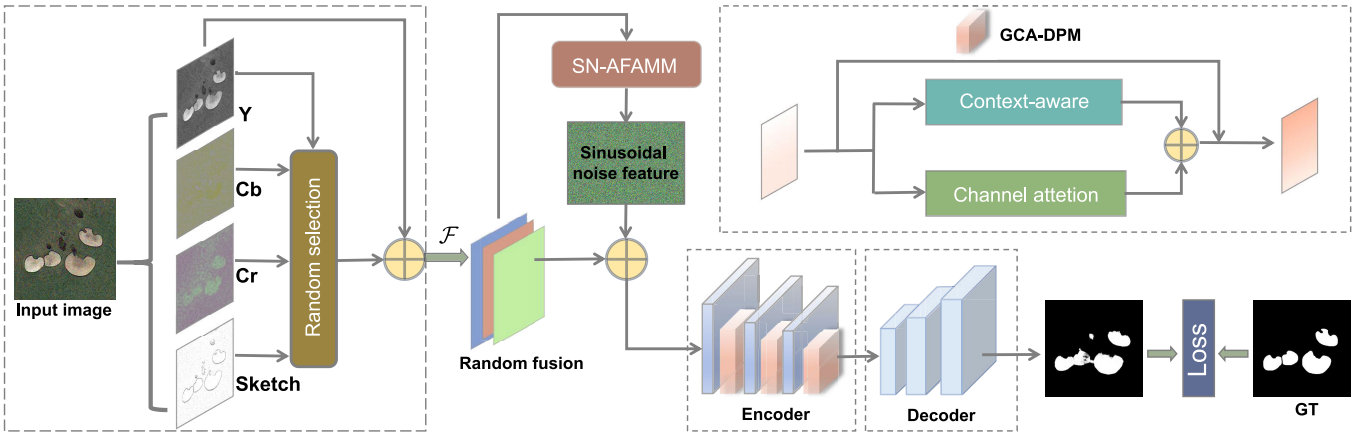


Fig. 2. Overall architecture of the proposed adversarial defense algorithm. First, the Y luminance channel of the adversarial sample is fixed. Then, two channels are randomly selected from Y, Cb, and Cr, and sketch image channels, and fused with the fixed Y channel to generate a new trichannel input. SN-AFAMM represents the sinusoidal noise-driven adversarial feature adaptive masking module. GCA-DPM denotes the global context-aware denoising perception module, which is embedded after each primary convolutional block in the encoder. The encoder and decoder are the main structures of the conventional SOD model.

training seeks to increase model robustness by dynamically generating adversarial examples during the training process and incorporating them into the empirical risk minimization to optimize model parameters [44], [45]. Defense strategies that improve model structures aim to enhance the model's robustness against adversarial samples by modifying or enhancing the model's architecture and components [46], [47], [48], [49]. For instance, Erichson et al. [46] propose a new activation function, JumpReLU, to enhance the model's representation learning ability. Li et al. [47] replace max-pooling, strided convolution, and average-pooling in CNNs with discrete wavelet transforms to enhance the model's adversarial robustness. Input transformation-based adversarial defense strategies involve a series of preprocessing steps that manipulate the image before passing it to the model, making it more difficult for the adversary's perturbations to take effect. These preprocessing steps include JPEG compression [50], bit reduction [51], application of filters [52], and purification of adversarial perturbations [53]. In addition, they may involve random transformations, such as rotation, translation, scaling, or color adjustments applied to input images [54], [55]. Another approach employs generative models to model adversarial perturbations and reconstruct natural samples [56], [57]. Furthermore, some defense algorithms aim not to eliminate adversarial patterns from images but to disrupt the perturbations by adding noise [13], [14], [58]. Most existing adversarial defense algorithms are designed for natural image classification tasks, but there is a significant lack of research on defense algorithms for the ORSI-SOD task.

III. Proposed Method

A. Architecture Overview

In this section, we introduce our proposed adversarial defense algorithm for the ORSI-SOD task, which significantly enhances the robustness of the original SOD models against adversarial samples. As illustrated in Fig. 2, we initially fix the Y luminance channel of the adversarial sample and then randomly select two channels from Y, Cb, and Cr, and the sketch image channels for channel fusion to generate a new trichannel image. This trichannel image is subsequently fed into a sinusoidal noise-driven adversarial feature adaptive masking module. This module dynamically adjusts the sinusoidal noise parameters to adaptively obscure the subtle features or gradient information that adversarial attacks rely on. In addition, it leverages the randomness of its sinusoidal noise to transform the input adversarial samples, making the model to obtain more diverse inputs during training. This reduces the overreliance on specific adversarial sample patterns, thereby enhancing the model's generalization capability. While sinusoidal noise plays a positive role in enhancing model robustness, it does not directly improve the model's ability to recognize the semantic information in images. Therefore, to further accentuate the saliency of the detected objects and bolster the model's semantic comprehension of images, a global context-aware denoising perception module (GCA-DPM) is embedded after each main convolutional block in the SOD network encoder. This module further weakens the negative impact of adversarial noise by reinforcing the network's understanding of global information and attention to key features, thus reducing the adversarial samples' likelihood of misleading the model and enhancing the accuracy of SOD. The collaborative effect of these components equips our adversarial defense algorithm with stronger robustness and higher detection performance under adversarial attacks, enabling it to handle complex ORSI-SOD tasks. In Section III-B, we will detail the working principles of each component.

B. Key Components of the Algorithm

1) Fixed Luminance Channel Multimodal Data Random Fusion Adversarial Defense Strategy (FLC-MDRF-ADS): ORSIs are known for their rich spectral information, which can effectively distinguish different types of ground objects. However, this characteristic can have a detrimental impact on the SOD tasks because SOD relies on the distinctiveness between the background and the target. The complex spectral properties in ORSIs may make it challenging to separate the target from the background with similar colors, thus reducing the detection

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TABLE IV
PERFORMANCE COMPARISON OF DEFENSE METHODS ON NATURAL SAMPLES

Dataset	Method	SeaNet			GateNet			PiCANet			AGNet		
		$F_\beta \uparrow$	MAE↓	$S_m \uparrow$	$F_\beta \uparrow$	MAE↓	$S_m \uparrow$	$F_\beta \uparrow$	MAE↓	$S_m \uparrow$	$F_\beta \uparrow$	MAE↓	$S_m \uparrow$
ORSSD	No Defense	0.8942	0.0105	0.9260	0.9257	0.0118	0.9211	0.8732	0.0199	0.8891	0.9123	0.0091	0.9395
	Smooth	0.8499	0.0229	0.8629	0.8908	0.0173	0.8952	0.8303	0.0317	0.8339	0.8847	0.0135	0.8798
	JPEG	0.8614	0.0283	0.8766	**0.9034**	**0.0125**	**0.9109**	0.8509	0.0266	0.8677	0.8894	0.0141	0.9142
	NRP	0.8433	0.0243	0.8614	0.8988	0.0170	0.8963	0.8481	0.0274	0.8652	0.9043	**0.0105**	**0.9270**
	RAN	0.8510	0.0260	0.8625	0.8807	0.0185	0.8829	0.8392	0.0259	0.8567	0.8821	0.0130	0.9159
	LeNo	0.8588	0.0254	0.8659	0.8873	0.0172	0.8989	0.8485	0.0265	0.8671	0.8891	0.0117	0.9097
	Ours	**0.8615**	**0.0201**	**0.8811**	0.8952	0.0152	0.9008	**0.8648**	**0.0207**	**0.8791**	**0.9085**	0.0111	0.9246
EORSSD	No Defense	0.8649	0.0073	0.9208	0.8698	0.0086	0.8737	0.8543	0.0130	0.8691	0.8891	0.0072	0.9285
	Smooth	0.8067	0.0146	0.8341	0.8676	0.0095	0.8678	0.8007	0.0183	0.8036	0.8501	0.0086	0.8798
	JPEG	0.8256	0.0196	0.8457	**0.8747**	**0.0072**	**0.8784**	0.8275	0.0163	0.8360	0.8536	0.0087	0.8830
	NRP	0.8107	0.0161	0.8358	0.8658	0.0087	0.8722	0.8223	0.0160	0.8340	0.8618	**0.0072**	**0.8883**
	RAN	0.8100	0.0184	0.8408	0.8648	0.0097	0.8633	0.8218	0.0168	0.8397	0.8532	0.0084	0.8719
	LeNo	0.8197	0.0171	0.8431	0.8683	0.0091	0.8725	0.8290	0.0165	0.8399	0.8574	0.0079	0.8744
	Ours	**0.8306**	**0.0129**	**0.8588**	0.8717	0.0080	0.8733	**0.8346**	**0.0128**	**0.8471**	**0.8682**	0.0076	0.8872

accuracy. Research [30] has shown that discarding some color information in training data can improve the model's robustness to color variations. In addition, Pestana et al. [31] point out that under the same attack conditions, the luminance channel in the YCbCr color space is more susceptible to adversarial noise than the chrominance channels. Based on these findings, we propose that when enhancing a model's adversarial robustness, the focus should be on processing the image's luminance channel rather than all channels. Enhancing the model's ability to recognize the robust features in the luminance channel and its adaptability to color variations can effectively improve the overall robustness. Furthermore, better-capturing shape or contour features not only improves the accuracy of ORSI-SOD in remote sensing images [59] but also enhances the model's adversarial robustness [34]. Although visible images and sketch images have different representations, they share the same fundamental structural information. Thus, integrating sketch images helps strengthen the model's learning of contour and structural features [60], [61], further aiding the model in distinguishing salient objects in complex backgrounds.

Based on the above analysis, this article proposes a fixed luminance channel multimodal data random fusion adversarial defense strategy. Specifically, we first process a color image into a sketch image. This process follows a common image processing method: converting the color image into a grayscale image, then inverting it, applying Gaussian blur, and finally merging it with the original grayscale image to create the sketch image. Subsequently, we transform the original color image from the RGB color space to the YCbCr color space. The relationship between the RGB and YCbCr color spaces is as follows:

$$\begin{aligned} Y &= 0 + 0.299R + 0.587G + 0.114B \\ C_b &= 128 - 0.168736R - 0.331264G + 0.5B \\ C_r &= 128 + 0.5R - 0.418688G - 0.081312B. \end{aligned} \tag{4}$$

As shown in Fig. 2, in the preprocessing stage, for each input remote sensing image $\mathbf{I}$, we first fix the luminance channel Y and then select two channels from Y, Cb, Cr, and sketch to merge through a randomization process. Specifically, we define a random variable P sampled from a uniform distribution $\mathcal{U}(0, 1)$), i.e., $P \sim \mathcal{U}(0, 1)$. The probability of transforming the input image into a new trichannel image $\mathbf{I}^*$ is determined by the random variable P. The probability of channel fusion occurring can be formally represented as $P_1 = \Pr(I \xrightarrow{\text{Fusion}} I^*) = P$, and the probability of no channel fusion (returning the original image) is $1 - P_1 = 1 - P$. In the case of channel fusion, we define a random fusion function $\mathcal{F}$, which randomly selects two channels from the channel set $C = \{Y, Cb, Cr, \text{Sketch}\}$ to fuse with the fixed Y channel, creating a new trichannel image. The random fusion function $\mathcal{F}$ has six possible fusion modes $\mathcal{F}_i, \quad i \in \{1, \ldots, 6\}$ which are $\mathcal{F}_1 = (Y, Y, Y)$, $\mathcal{F}_2 = (Y, Cb, \text{Sketch})$, $\mathcal{F}_3 = (Y, Cr, \text{Sketch})$, $\mathcal{F}_4 = (Y, Y, \text{Sketch})$, $\mathcal{F}_5 = (Y, Y, Cb)$, and $\mathcal{F}_6 = (Y, Y, Cr)$. The selection of each fusion mode $\mathcal{F}_i$ follows the following conditional probability distribution:

$$\Pr(\mathcal{F}_i | P_1) = \frac{P_1}{6} \quad \forall i \in \{1, \ldots, 6\}. \tag{5}$$

The fixed luminance channel multimodal data random fusion adversarial defense strategy that we propose exhibits multiple advantages. First, in the complex spectral environment of ORSIs, perturbations tend to affect the luminance information, and the luminance channel Y usually contains the key structural information of the image. Therefore, by fixing the Y channel and randomly discarding the color channels, the model's ability to recognize robust features in the Y channel is enhanced. This helps maintain or restore the structural features of the image and reduces the model's reliance on color features, thereby improving the model's robustness to color deviations in complex spectral backgrounds. Second, this strategy incorporates sketch images to fully utilize the contour and structural information of the image while suppressing excessive details. The increased contrast in sketch images helps the model better understand the semantic divisions of the image and guides the model to focus on the learning key shape and contour features, thereby reducing its reliance on local details. Moreover, fusing sketch images with some channels of natural samples as input provides complementary structural information. This not only helps improve the model's ability to judge overall structures but also overcomes the issue of adversarial attacks confusing the model by perturbing detail features. Finally, this strategy exposes the model to a variety

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TABLE V
INDIVIDUAL CONTRIBUTIONS OF THE THREE COMPONENTS

Component			Metric		
FLC	SN	GCA	$F_\beta \uparrow$	MAE$\downarrow$	$S_m \uparrow$
×	×	×	0.1496	0.0916	0.4908
✓	×	×	0.6441	0.0498	0.6812
×	✓	×	0.6033	0.0511	0.6501
×	×	✓	0.5891	0.0582	0.6278
×	✓	✓	0.7385	0.0214	0.7918
✓	×	✓	0.7282	0.0191	0.7912
✓	✓	×	0.6825	0.0241	0.7561
✓	✓	✓	**0.7726**	**0.0157**	**0.8153**

of inputs and uncertainties during the training process. This diversity and uncertainty drive the model to learn more robust and generalizable feature representations. This allows the model to maintain stability and discriminative capability when facing changing input conditions, thereby exhibiting higher performance in adversarial attack scenarios.

2) Sinusoidal Noise-Driven Adversarial Feature Adaptive Masking Module (SN-AFAMM): After the original image is preprocessed and transformed into tensor form, it is denoted as input feature maps x. Let $F(x)$ be the original SOD model, and our SN-AFAMM is defined as SN-AFAMM$(x) = \alpha \cdot \sin(\beta x)$. As shown in Fig. 2, we add the sinusoidal noise feature through SN-AFAMM to the original feature x and denote the resulting feature as $T(x)$, i.e., $T(x) = x + \alpha \cdot \sin(\beta x)$. During the module's initialization, the parameter α is set to a value close to 0, while β is set to a significantly larger value greater than 0. This setting makes the feature $T(x)$ approximate the original input x, i.e., $T(x) \approx x$. Let the SOD model with this module be denoted by $F(T(x))$, then $F(T(x)) \approx F(x)$.

In scenarios of white-box attacks, where $\mathcal{L}$ represents the binary cross-entropy loss function, the backward propagation of the loss function for the model $F(T(x))$ is as follows:

$$\left(\frac{\partial \mathcal{L}}{\partial x}\right)_{\text{SN}} = \frac{\partial \mathcal{L}}{\partial F} \cdot \frac{\partial F}{\partial T} \cdot \frac{\partial T}{\partial x} = \frac{\partial \mathcal{L}}{\partial F} \cdot \frac{\partial F}{\partial T} \cdot (1 + \alpha\beta \cos(\beta x)). \tag{6}$$

For the original model $F(x)$ without this module, the backward propagation of the loss function is

$$\left(\frac{\partial \mathcal{L}}{\partial x}\right)_{\text{original}} = \frac{\partial \mathcal{L}}{\partial F} \cdot \frac{\partial F}{\partial x}. \tag{7}$$

Since $T(x) \approx x$, we have

$$\frac{\partial F}{\partial T} = \frac{\partial F}{\partial x}. \tag{8}$$

Combining the above equations, we can obtain

$$\left(\frac{\partial \mathcal{L}}{\partial x}\right)_{\text{SN}} \approx \left(\frac{\partial \mathcal{L}}{\partial x}\right)_{\text{original}} \cdot (1 + \alpha\beta \cos(\beta x)). \tag{9}$$

Given that α is close to 0 and β is much greater than 0, the product $\alpha\beta$ is much greater (or less) than 1. In this case, when adversarial samples enter the module, the gradients undergo complex nonlinear transformations. This effectively masks the adversarial sample's original disruptive noise, severely hindering the attacker's optimization process based on the source model's gradients in white-box attacks. Consequently, this increases the difficulty of adversarial sample attacks, thereby enhancing the model's adversarial robustness.

In black-box attack scenarios, adding the sinusoidal spatial noise (α and β) generated by SN-AFAMM to the original features of the adversarial sample changes the distribution of input data and increases data uncertainty. This process leads to a blurring of adversarial samples that were originally disruptive to the model. Such blurring helps to conceal the adversarial samples' disruptive features, thereby reducing their disruptive effect on the model. Furthermore, different models may display varied sensitivities when confronted with adversarial attacks. The learnable parameters in SN-AFAMM not only provide it with the ability to make personalized adjustments between different models but also allow it to adaptively update during training, thereby maximizing the adversarial robustness of each model. On the other hand, the randomness in the sinusoidal noise can serve as a form for regularization, preventing the model from over-optimizing for specific adversarial sample features. In terms of optimization objectives, the noise forces the model to adapt to a broader range of input perturbations, not just the existing adversarial disturbances. This approach helps to avoid the model's overreliance on specific adversarial sample features, thereby enhancing its generalization capabilities against unknown attacks. The specific parameter settings and optimization process of SN-AFAMM will be detailed in the experimental section.

3) Global Context-Aware Denoising Perception Module (GCA-DPM): The introduced sinusoidal noise in SN-AFAMM is considered a means for enhancing robustness. While it helps improving the model's generalization capabilities, it does not directly enhance the model's feature representation and discrimination abilities for targets. Therefore, we propose a GCA-DPM and embed it after each main convolution block of the SOD network encoder. This module is designed to further mitigate the negative effects of adversarial perturbations and continuously strengthen the deep feature expression, thereby enhancing the model's semantic understanding of images and reducing the misleading effects of adversarial samples. The collaboration of these two modules helps balancing the model's generalization capabilities and its understanding of image content. They complement each other, making our defense algorithm with greater robustness in handling adversarial samples. The structural design of the GCA-DPM is illustrated in Fig. 3.

Consider a feature map denoted by $\mathbf{A} \in \mathbb{R}^{C\times H\times W}$, which is input to GCA-DPM. In branch 1, we reshape the feature map $\mathbf{A}$ to $\mathbb{R}^{C\times HW}$ and then perform matrix multiplication between $\mathbf{A}$ and its transpose $\mathbf{A}^{\mathbf{T}}$. After applying a softmax layer, we obtain the channel attention map $\mathbf{X} \in \mathbb{R}^{C\times C}$. Subsequently, we perform matrix multiplication between $\mathbf{X}$ and $\mathbf{A}$, and then reshape the result back to the original feature map size, resulting in the output $\mathbf{D} \in \mathbb{R}^{C\times H\times W}$. This branch employs a channel attention mechanism to dynamically adjust the weights between different channels based on the input data, enhancing or weakening the relative importance of different

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TABLE VI
COMBINED CONTRIBUTIONS OF THREE COMPONENTS TO SOD MODEL PERFORMANCE

Component			SeaNet			GateNet			PiCANet			AGNet		
FLC	SN	GCA	$F_\beta \uparrow$	MAE↓	$S_m \uparrow$	$F_\beta \uparrow$	MAE↓	$S_m \uparrow$	$F_\beta \uparrow$	MAE↓	$S_m \uparrow$	$F_\beta \uparrow$	MAE↓	$S_m \uparrow$
×	×	×	0.1496	0.0916	0.4908	0.6321	0.0895	0.6920	0.5995	0.0641	0.6211	0.6694	0.0324	0.7472
×	✓	✓	0.7385	0.0214	0.7918	0.8011	0.0122	0.8204	0.7663	0.0185	0.8056	0.8232	0.0096	0.8644
✓	×	✓	0.7282	0.0191	0.7912	0.7961	0.0136	0.8088	0.7714	0.0188	0.8094	0.8268	0.0093	0.8619
✓	✓	×	0.6825	0.0241	0.7561	0.7824	0.0137	0.8121	0.7404	0.0192	0.7951	0.7891	0.0123	0.8373
✓	✓	✓	**0.7726**	**0.0157**	**0.8153**	**0.8173**	**0.0116**	**0.8346**	**0.7819**	**0.0167**	**0.8131**	**0.8317**	**0.0091**	**0.8663**

TABLE VII
TRAINING RESULTS OF LEARNABLE PARAMETERS IN SN-AFAMM

Attack	Parameter	SeaNet	GateNet	PiCANet	AGNet
FGSM	α	1.87e-03	-7.23e-04	-1.06e-03	1.91e-02
	β	1.08e+21	8.47e+19	9.20e+17	1.81e+22
PGD	α	2.71e-03	1.28e-04	-2.45e-03	1.80e-02
	β	1.21e+21	8.55e+19	9.40e+17	1.49e+22
APGD	α	3.42e-03	-1.75e-04	-5.32e-03	-1.57e-02
	β	1.45e+21	8.21e+19	9.72e+17	1.13e+22

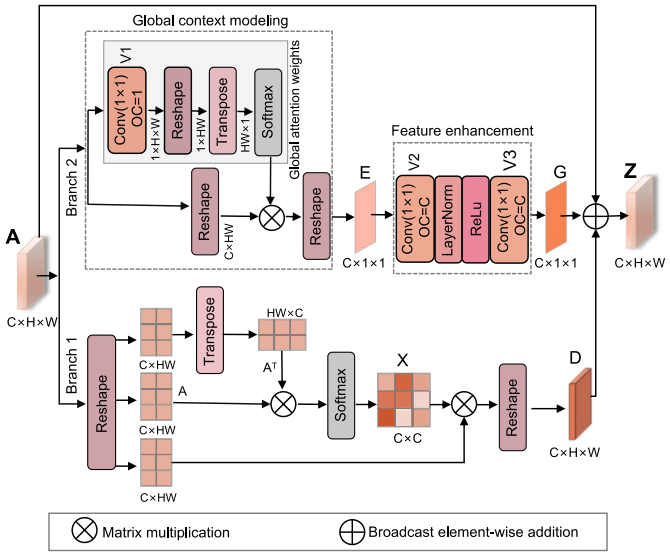


Fig. 3. Global context-aware denoising perception module.

channel features in the feature map, thereby improving the model's perception of varying features.

In branch 2, three 1 × 1 convolutions represent the linear transformation matrices **V1**, **V2**, and **V3**. The output channel (OC) number of **V1** is 1, while the OC numbers of **V2** and **V3** are equal to the input channel number. First, we model the global context of the input feature map **A** to get the feature association matrix $\mathbf{E} \in \mathbb{R}^{C\times1\times1}$

$$E = \sum_{i=1}^{N_p} \frac{\exp(V_1 \cdot A_i)}{\sum_{j=1}^{N_p} \exp\left(V_1 \cdot A_j\right)} A_i \tag{10}$$

where N_P denotes the number of positions in the feature map, and $N_P = H \times W$. A_i is the feature vector at a position in the feature map, and $((\exp(V_1 \cdot A_i))/(\sum_{j=1}^{N_p} \exp(V_1 \cdot A_j)))$ represents the global attention weights obtained after applying the softmax layer. Each position in **E** contains a context vector

TABLE VIII
COMPUTATIONAL COMPLEXITY AND LATENCY PERFORMANCE OF EACH MODEL

Model	Metric	Baseline	Baseline + Ours
SeaNet	FLOPs(G)	1.8094	1.8106
	Latency(ms)	31.6889	32.1590
GateNet	FLOPs(G)	162.1948	162.2287
	Latency(ms)	20.8856	21.1670
PiCANet	FLOPs(G)	59.7860	59.8052
	Latency(ms)	117.4436	117.7581
AGNet	FLOPs(G)	9.6783	9.6809
	Latency(ms)	83.3621	84.0315

based on the input feature map and global attention weights, which adjust the importance of the original features in the global context. This enables the model to effectively capture dependencies between distant regions in the image, thereby enhancing the understanding of the overall content. Subsequently, we process and map the contextual information in the feature association matrix **E** along the channel dimension to obtain the context-enhanced feature representation $\mathbf{G} \in \mathbb{R}^{C\times1\times1}$

$$G = V_3\,\mathrm{ReLU}(\mathrm{LN}(V_2 E)). \tag{11}$$

This context-enhanced feature representation assigns a more robust and effective feature expression to each spatial position, improving the model's understanding and utilization of contextual information, thereby enhancing the model's global perception ability. Finally, the context vectors of the enhanced features, along with global channel information, are fused with the original input features to obtain the final global feature representation $\mathbf{Z} \in \mathbb{R}^{C\times H\times W}$.

In ORSIs, small targets are commonly present, and the edge features of the objects in images tend to be blurred due to the influence of the natural environment, which poses a challenge for SOD models. This module, by integrating context-aware mechanisms and channel attention mechanisms not only enhances the saliency of target features to improve the distinguishability of targets against complex backgrounds but also helps the model focus on key features to reduce the impact of perturbative features. The SN-AFAMM enhances the model's generalization to different adversarial samples by injecting random noise. Concurrently, the GCA-DPM leverages global contextual information to strengthen feature expression and judgment, enabling the model to suppress the interference of adversarial noise and thus maintain its ability to detect salient targets effectively. These two components

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Algorithm 1 Implementation Process of the Proposed Adversarial Defense Algorithm

Input: Adversarial optical remote sensing image I and its corresponding ground truth.
Output: Robust SOD model.

```
Initialize the SOD model.
Apply FLC-MDRF-ADS to I:
    Generate a sketch image from I.
    Convert I from RGB to YCbCr color space.
    Create a new tri-channel image I* according to the channel fusion function F.
Transform I* into tensor x.
Initialize the parameters α to be close to 0 and β to be significantly greater than 0 for the SN-AFAMM.
for i = 1 to iterations do
    Apply SN-AFAMM to input feature map x to obtain feature T(x) = x + α · sin(β · x).
    Forward pass T(x) through the SOD model;
    for j = 1 to the number of main convolutional blocks in the encoder do
        Extract features from the j-th convolutional block to obtain a feature map A.
        Apply the GCA-DPM to A:
            Apply channel attention and global context-aware processing to enhance feature map A, obtain Z.
        Forward pass Z through the next convolutional block.
    end for
    Forward pass the obtained features through the decoder to get the saliency map.
    Calculate the loss between the saliency map and the ground truth.
    Backpropagate the loss and update the model parameters.
end for
```

work together through a dual strategy of joint generalization enhancement and feature expression optimization, further providing the model with an efficient adversarial defense mechanism. The implementation process of our proposed adversarial defense algorithm is shown in Algorithm 1.

IV. Experimental Results

A. Experimental Setup

1) Datasets: The ORSSD dataset [36] comprises 800 ORSIs along with their corresponding pixel-level annotations. This dataset is divided into two parts: a training set with 600 images and a testing set with 200 images. The EORSSD dataset [8] is an extension of the ORSSD dataset, adding an additional 1200 ORSIs for a total of 2000 ORSIs. These images are split into a training set of 1400 images and a testing set of 600 images. These two datasets for SOD in remote sensing imagery are highly challenging due to differences in spatial resolution, complex and diverse backgrounds, various types of targets, and variations in the number and size of salient objects. For the experimental portion of this article, we use the training sets of ORSSD and EORSSD for model training and evaluate the SOD performance on the test sets of both ORSSD and EORSSD.

2) Evaluation Metrics: We evaluate the performance of remote sensing SOD using three metrics: F-measure (F_β), mean absolute error (MAE) score, and S-measure (S_m). F-measure is defined as the weighted harmonic mean of precision and recall, serving as a comprehensive metric [62]. Its mathematical expression is as follows:

$$F_\beta = \frac{\left(1+\beta^2\right)\text{Precision} \times \text{Recall}}{\beta^2 \times \text{Precision} + \text{Recall}} \tag{12}$$

where β^2 is set to 0.3 to emphasize the model's performance in terms of accuracy [63]. A higher F_β value indicates better overall performance of the model. MAE measures pixel-level differences between the saliency map S and the ground truth G [64] and is calculated as follows:

$$\text{MAE} = \frac{1}{W \times H} \sum_{m=1}^{W} \sum_{n=1}^{H} |S(m,n) - G(m,n)| \tag{13}$$

where H and W represent the height and width of the input image, respectively. A smaller MAE value indicates the predicted image is closer to the actual ground truth. S-measure evaluates the structural similarity between the saliency map and the ground truth [65], with its computational formula as follows:

$$S_m = \alpha \times S_o + (1-\alpha) \times S_r \tag{14}$$

where S_r and S_o represent the region similarity and object similarity, respectively. A higher S_m value denotes superior performance.

3) Implementation Details: In this study, we employ three typical gradient-based adversarial attack methods, FGSM, PGD, and APGD to attack the pretrained ORSI-SOD model SeaNet [11] in order to generate adversarial samples. Specifically, we set the attack step size of FGSM to 0.3, and the attack step size of PGD to 0.04 with ten iterations. For the APGD attack, we initialize the attack step size to 0.04 and limit the maximum number of iterations to 30. Using these three attack methods, we construct three adversarial sample datasets for ORSIs in the SOD task and evaluate the robustness of other SOD models based on these datasets. It is noteworthy that our evaluation process includes a white-box attack on the SeaNet model, along with black-box attacks on the GateNet [6], PiCANet [5], and AGNet [10] models. The training parameter settings for each SOD model are shown in Table I. The experiments in this study are implemented on the PyTorch platform and executed on a workstation with two NVIDIA A100 (40 GB) GPUs.

B. Sensitivity of the Models to Adversarial Attacks

We evaluate the performance of four state-of-the-art SOD models (SeaNet, GateNet, PiCANet, and AGNet) on both natural images and adversarial samples. These adversarial samples are generated based on the pretrained SeaNet model. Table II presents the quantitative results of these models under FGSM, PGD, and APGD adversarial samples, showing a significant performance drop for all models when facing

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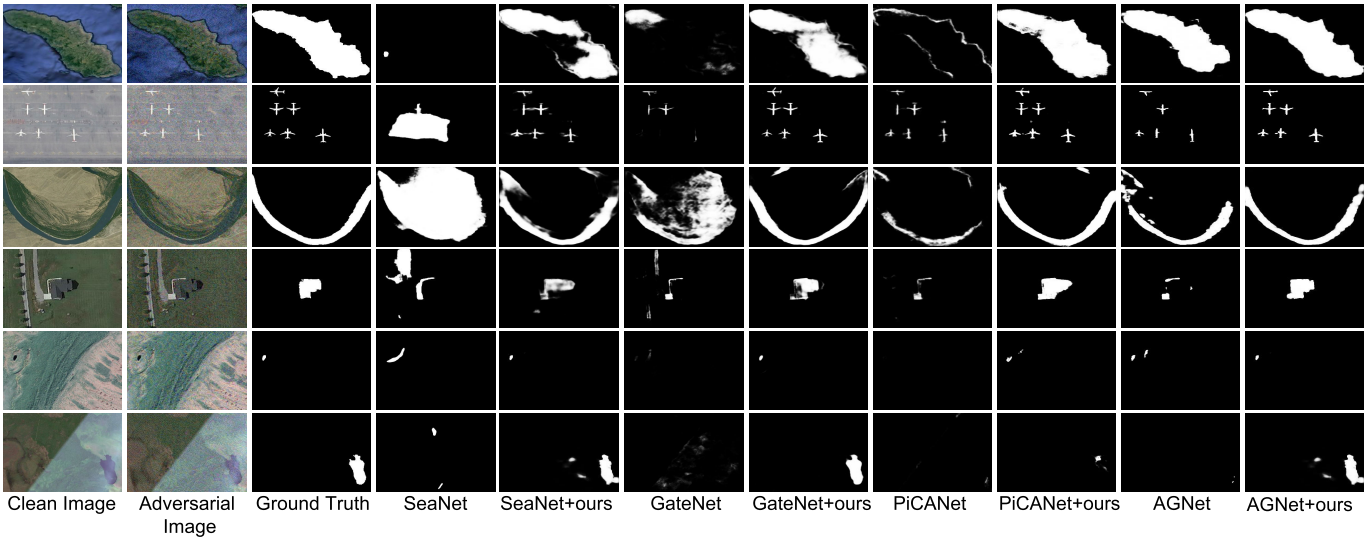


Fig. 4. Robustness of the proposed defense algorithm. The leftmost column shows the natural samples, the second column displays the corresponding adversarial samples, and the third column presents the ground truth of salient objects. N + ours denotes the model N combined with our methods. All SOD models are tested on PGD adversarial samples. As depicted in this figure, our proposed algorithm significantly enhances the detection performance of the SeaNet, GateNet, PiCANet, and AGNet models against adversarial samples.

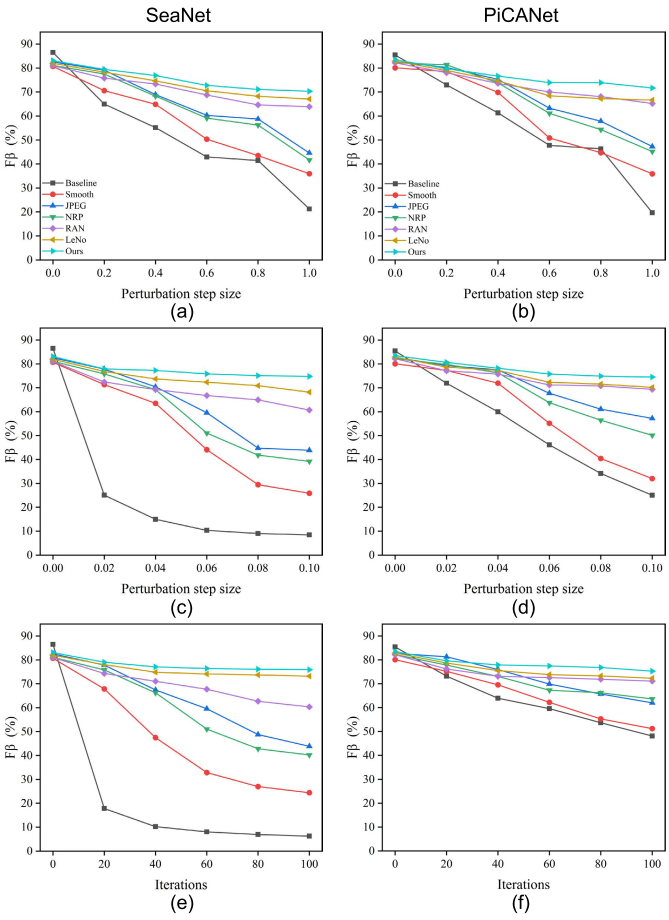


Fig. 5. Performance of the SeaNet model and the PiCANet model under different attack budgets. (a), (c), and (e) Performance of the SeaNet model under FGSM, PGD, and APGD attacks, respectively. (b), (d), and (f) Performance of the PiCANet model under FGSM, PGD, and APGD attacks, respectively.

adversarial examples. It can be seen that SeaNet suffers the most significant performance loss under adversarial samples, likely due to the adversarial samples being crafted based on SeaNet's own architecture. Moreover, other models are also affected to varying degrees, possibly related to the similarity between their architectures and the pretrained models used for generating adversarial samples. Fig. 1 displays the saliency map predictions of different SOD models under PGD adversarial attacks, demonstrating that the prediction quality of all models significantly deteriorates, exhibiting noticeable inaccuracies.

C. Performance Analysis

To evaluate the performance of our proposed defense algorithm, we conduct a series of comparative experiments with other classic defense methods, specifically including spatial smoothing [52], JPEG compression [50], neural representation purifier (NRP) [53], RAN [49], and LeNo [14]. Table III presents how different defense methods affect the performance of various detection models against adversarial samples from FGSM, PGD, and APGD attacks, respectively. As shown in the table, when our defense method is deployed across different models, all models exhibit significant performance improvements. From F_β, MAE, and S_m metrics of the models, it can be observed that our defense algorithm demonstrates superior performance in maintaining high detection accuracy, low error rates, and high similarity matching, achieving the best results in most cases. This strongly validates the significant advantages of our proposed defense method in enhancing the adversarial robustness of SOD models, highlighting the effectiveness and practical value of our approach. Fig. 4 provides some qualitative comparisons of the robustness of our proposed defense algorithm.

The comparative performance of defense methods on natural samples is presented in Table IV. The experimental results demonstrate that our defense method not only exhibits exceptional adversarial robustness on adversarial samples but

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also maintains relatively high detection accuracy on natural samples. Specifically, on the GateNet model, our defense method, although slightly underperforming compared with methods based on JPEG compression, essentially achieves suboptimal performance. In comparison with the SeaNet, PiCANet, and AGNet models, other defense methods cause more significant performance degradation, while our method typically shows relatively smaller declines in performance, reflecting good stability. The experimental results indicate that although our method is not the best in certain scenarios, overall, it achieves a good balance between enhancing the robustness against adversarial samples and maintaining the accuracy of detecting natural samples, demonstrating strong potential for application.

To further evaluate the impact of different attack budgets on the robustness of the proposed defense algorithm, we select the SeaNet model under white-box attacks and the PiCANet model under black-box attacks as the subjects of study. The experiments are conducted on the EORSSD dataset, and the model performance is evaluated using the F_β metric. The specific attack settings are as follows: for FGSM attacks, the perturbation step size starts at 0.2 and increases by intervals of 0.2–1; for PGD attacks, the number of iterations is fixed at 10, with the perturbation step size starting at 0.02 and increasing by intervals of 0.02 up to 0.1; for APGD attacks, we adjust the attack budget by changing the number of iterations, initializing the perturbation step size at 0.04, and the number of iterations starts at 20 and increases by intervals of 20–100. The experimental results in Fig. 5 demonstrate that as the attack budget increases, the model's robustness faces more severe challenges. Specifically, PiCANet exhibits better robustness than SeaNet, which may be attributed to the limited understanding of the model by the attacker in a black-box environment. In addition, regardless of whether in white-box or black-box attack environments, when the perturbation budget is low, the performance differences between different defense methods are small, and methods like JPEG or NRP sometimes achieve optimal performance, while the performance of smooth is also close. However, as the perturbation budget increases, the performance of these preprocessing methods significantly declines. This phenomenon may be due to the fact that when facing minor perturbations, Smooth, JPEG, and NRP can effectively eliminate slight adversarial perturbations by directly processing input data without significantly altering the content of the original image. However, as the perturbation intensity increases, these methods either struggle to eliminate more powerful and complex adversarial perturbations or may degrade the performance by overly processing and altering the original features of the image. In contrast, RAN, LeNo, and our method demonstrate relatively stable performance. These methods enhance the model's intrinsic robustness to improve its ability to recognize and handle adversarial perturbations, enabling the model to resist more powerful adversarial attacks. It is worth noting that our defense algorithm exhibits more stable and superior defense capabilities under different attack budgets, further validating the effectiveness of the proposed algorithm in addressing adversarial sample challenges in complex ORSI-SOD application scenarios.

D. Ablation Studies

The key components of our proposed defense algorithm include the FLC-MDRF-ADS, the SN-AFAMM, and the GCA-DPM. To assess the contributions of the key components of our proposed method in enhancing adversarial robustness, we conduct a comprehensive ablation study on PGD adversarial samples in EORSSD. It is worth noting that inspired by [66], we initialize the learnable parameters α and β in the SN module to $1e^{-8}$ and $1e^{20}$. The SN module uses the same optimizer as the baseline model, with a learning rate set to 0.01.

1) Individual Contributions of the Three Components: To accurately evaluate the independent and collaborative effects of the key components, we select SeaNet as the baseline model and augment it with different components to assess their individual and combined impacts. Table V presents the quantitative results of this experiment. The experimental results indicate that when FLC, SN, and GCA modules are added individually, all three performance metrics show improvements. When any two modules are combined, the performance improvement is more pronounced, indicating a positive mutual enhancement effect between the modules. The most significant performance improvement is achieved when we simultaneously introduce all the three modules into the SOD model. The improvements in F_β, MAE, and S_m values exceed any other combinations of modules, demonstrating the strong effectiveness of the comprehensive defense strategy formed by the three modules in enhancing the model's robustness against adversarial samples. This result not only validates the effectiveness of individual components in enhancing adversarial robustness but also highlights the synergistic effects between components that can achieve optimal defense results across a wider range of performance metrics.

2) Combined Contributions of the Three Components: To evaluate the combined contributions of the three components to SOD model performance, we select SeaNet, GateNet, PiCANet, and AGNet as baseline models and perform ablation studies on each model to assess the contributions of any two combinations of our three components. The experimental results are presented in Table VI. The experimental results show that in the baseline models without deploying any defense modules, the performance of all SOD models significantly deteriorates on adversarial samples. When we introduce defense modules in pairwise combinations into the models, the performance of the models is significantly improved. It is worth noting that in the combinations of SN with GCA and FLC with GCA, the performance of SOD models is generally superior to the case, where FLC and SN are combined. This phenomenon suggests that GCA plays a crucial role in enhancing the saliency features of objects by integrating global information and suppressing the interference of adversarial noise on the model, thereby improving the model's robustness. When all three modules are combined, the model achieves the maximum performance improvement. This result confirms that our proposed defense algorithm benefits from all the proposed defense modules.

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We deploy the proposed adversarial defense algorithm to different SOD models for training. After model convergence, the output results of the learnable parameters α and β in the sinusoidal noise-driven adversarial feature adaptive masking module on different adversarial samples are presented in Table VII. The experimental results indicate that different models exhibit distinct parameter responses under the three types of attacks, illustrating that the SN-AFAMM enables models to dynamically adjust their parameters based on their intrinsic architecture and training data characteristics, thus allowing them to adapt to adversarial attacks. Notably, for the same model facing different adversarial attacks, the exponential magnitudes of parameters α and β exhibit relative consistency, while their specific values show slight variations. This phenomenon may be closely related to the uniformity of the model architecture, meaning that the same model architecture has inherent constraints on the range of parameter changes under different adversarial attacks, maintaining the stability of exponential magnitudes. However, despite the consistent model architecture, subtle adjustments during dynamic training lead to parameter variations when facing different adversarial samples, aiming to maximize the model's responsiveness to each specific adversarial sample type. Therefore, the numerical differences in parameters α and β under different attack methods actually reflect the model's fine-tuning ability during dynamic learning and adaptation. This ability to fine-tune numerical values while maintaining exponential consistency highlights the effectiveness of SN-AFAMM in enhancing the adversarial robustness of neural network models.

Considering the efficiency requirements of ORSI-SOD in practical applications, we adopt the floating point operations per second (FLOPs) and latency as metrics to evaluate the impact of the proposed defense algorithm on the model's computational complexity and response time. Specifically, for latency evaluation, we perform inference on the EORSSD test set using each SOD model. Each model undergoes ten tests, and the average latency value across these tests is considered the final latency value of the model. The experimental results in Table VIII show that our algorithm incurs only very slight increases in the computational complexity and response time of all models. This indicates that the proposed defense algorithm can enhance model robustness while ensuring computational efficiency and responsiveness, demonstrating good efficiency and practicality.

V. Conclusion

The adversarial robustness of ORSI-SOD models is of crucial significance for ensuring the reliability of remote sensing images in practical applications. However, existing defense research has rarely focused on the ORSI-SOD task, particularly lacking effective defense solutions tailored to the unique challenges of this task. In this article, we propose a robust adversarial defense algorithm for ORSI-SOD for the first time. Initially, to mitigate the interference from complex backgrounds, we propose a fixed luminance channel multimodal data random fusion adversarial defense strategy, which enhances the model's adaptability to color and contour changes in salient targets. Second, to enhance the model's generalization ability and reduce the effectiveness of adversarial attacks, we propose a sinusoidal noise-driven adversarial feature adaptive masking module. This module adaptively disrupts the interfering features of adversarial samples and exposes the model to a more diverse data distribution. Finally, to enhance the saliency of target features, we propose a GCA-DPM to improve the model's recognition ability and further suppress the negative impact of adversarial noise on the model. These three modules collaborate to build a robust adversarial defense mechanism. Extensive experimental results demonstrate that the adversarial defense algorithm presented in this article can significantly improve the adversarial robustness of various advanced SOD models.

However, this article also has certain limitations. For instance, it does not incorporate adversarial training into the optimization process of the algorithm to further enhance the model's potential to counter adversarial attacks. In addition, the current evaluation of the algorithm's performance only considers three types of classical adversarial attacks, without covering more complex adversarial attack methods. In future work, we plan to address these limitations by further optimizing and validating the universality and effectiveness of the proposed defense algorithm, aiming to provide safer and more reliable protection for the analysis and processing of remote sensing images.

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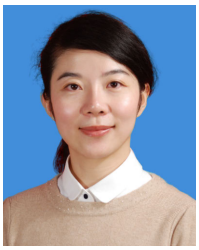
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